

Pass-Through and Incidence under Product Returns

Draft notes

July 19, 2026

Abstract

Pass-through and incidence when a seller faces product returns. The aggregate return rate $r(p)$ —the share of units sold at price p that are returned—rises with p . Two per-unit costs enter: production cost c (all units) and handling cost h (returned units only). Results are derived for perfect competition and monopoly.

1 Notation

- $Q = D(p)$: demand at list price p , $D' < 0$; $\varepsilon_D \equiv -pD'/D > 0$ (demand elasticity w.r.t. p).
- $Q = S(\mu)$: supply at effective margin μ , $S' > 0$.
- $r(p) \in (0, 1)$: aggregate (conditional) return rate at p ; $r'(p) > 0$.
- c : per-unit production cost; h : per-unit handling cost on returns.
- $\mu(p, c, h)$: expected profit per unit sold (effective margin).
- $\mu_p \equiv \partial\mu/\partial p$, $\mu_c \equiv \partial\mu/\partial c$, $\mu_h \equiv \partial\mu/\partial h$.
- $\rho_c \equiv dp/dc$, $\rho_h \equiv dp/dh$: pass-through into p .
- I : incidence ratio $|\Delta CS|/|\Delta PS|$ (PC) or $|\Delta CS|/|\Delta\pi|$ (monopoly).

2 Perfect competition with product returns

Consumers pay list price p . Producers sell Q units; share $r(p)$ is returned.

2.1 Model

Per unit sold at p : a kept unit yields $(p - c)$; a returned unit yields $(-c - h)$. Expected margin per unit sold is

$$\mu(p, c, h) \equiv p - c - r(p)(p + h),$$

so

$$\mu_p = 1 - r(p) - r'(p)(p + h), \quad \mu_c = -1, \quad \mu_h = -r(p), \quad \mu_{pc} = 0, \quad \mu_{ph} = -r'(p).$$

Supply is $Q = S(\mu)$ with $S' > 0$. Market clearing:

$$D(p) = S(\mu(p, c, h)). \tag{1}$$

Here Q is **gross purchases** at list price p : $D(p)$ is checkout volume and share $r(p)$ return ex post.

2.2 Pass-through of c and h

Market clearing $G(p, c, h) \equiv D(p) - S(\mu(p, c, h)) = 0$ pins down $p(c, h)$. Totally differentiate w.r.t. $x \in \{c, h\}$:

$$G_p \rho_x + G_x = 0, \quad G_p = D'(p) - S'(\mu) \mu_p, \quad G_x = -S'(\mu) \mu_x,$$

since D does not depend on (c, h) directly. Hence

$$\rho_x = \frac{S'(\mu) \mu_x}{D'(p) - S'(\mu) \mu_p}. \quad (2)$$

In Weyl and Fabinger (2013) notation, $\varepsilon_D \equiv -pD'/Q$ and $\varepsilon_S \equiv (p/Q) dQ/dp$. From (1), supply is indexed by μ and $dQ/dp = S'(\mu) \mu_p$, so $\varepsilon_S = pS'(\mu) \mu_p / Q$ (reduces to $pS'(p-c)/Q$ when $r \equiv 0$). Substituting $D' = -\varepsilon_D Q/p$ into (2):

$$\rho_x = \frac{pS'(\mu) \mu_x / Q}{\varepsilon_D + \varepsilon_S} = \frac{\varepsilon_S \mu_x}{\mu_p (\varepsilon_D + \varepsilon_S)}. \quad (3)$$

Production cost c . $\mu_c = -1$, $\mu_{pc} = 0$. From (3):

$$\rho_c = \frac{\varepsilon_S}{\mu_p (\varepsilon_S + \varepsilon_D)}. \quad (4)$$

When $r \equiv 0$ ($\mu_p = 1$), $\rho_c = \varepsilon_S / (\varepsilon_S + \varepsilon_D)$.

Handling cost h . $\mu_h = -r(p)$. From (3):

$$\rho_h = \frac{\varepsilon_S r(p)}{\mu_p (\varepsilon_S + \varepsilon_D)}, \quad \frac{\rho_h}{\rho_c} = r(p). \quad (5)$$

Example: $r(p) = \alpha$, $D(p) = a - bp$, $S(\mu) = d + e\mu$. Clearing gives $a - bp = d + e(p - c - \alpha(p + h))$, i.e. $a - d - c - \alpha h = [b + e(1 - \alpha)]p$. Differentiate w.r.t. c : $0 = [b + e(1 - \alpha)]\rho_c - e$, so $\rho_c = e/[b + e(1 - \alpha)]$. Differentiate w.r.t. h : $-\alpha = [b + e(1 - \alpha)]\rho_h$, so $\rho_h = \alpha e/[b + e(1 - \alpha)] = \alpha \rho_c$.

2.3 Incidence

Producer surplus is $PS(\mu) = \int_0^\mu S(x) dx$, so $PS'(\mu) = Q$. Consumer surplus satisfies $dCS = -Q dp$. Along the equilibrium path,

$$\frac{dCS}{dx} = -Q \rho_x, \quad \frac{dPS}{dx} = Q(\mu_p \rho_x + \mu_x), \quad x \in \{c, h\}. \quad (6)$$

Production cost.

$$I_c \equiv \frac{|dCS/dc|}{|dPS/dc|} = \frac{\rho_c}{1 - \mu_p \rho_c}. \quad (7)$$

When $r \equiv 0$ ($\mu_p = 1$), $I_c = \rho_c / (1 - \rho_c)$.

Handling cost. Using $\rho_h = r \rho_c$,

$$I_h = \frac{\rho_h}{r - \mu_p \rho_h} = \frac{\rho_c}{1 - \mu_p \rho_c} = I_c. \quad (8)$$

Consumers bear the same fraction of a handling shock as of a production shock.

2.4 Summary: perfect competition

Object	Formula
Margin	$\mu = p - c - r(p)(p + h)$
ρ_c	$\varepsilon_S / [\mu_p(\varepsilon_S + \varepsilon_D)]$
ρ_h	$r \rho_c$
I_c, I_h	$\rho_c / (1 - \mu_p \rho_c)$

3 Monopoly with product returns

A monopolist chooses p to maximize $\pi(p) = D(p) \mu(p, c, h)$ with $\mu = p - c - r(p)(p + h)$ and $\mu_p = 1 - r - r'(p)(p + h)$.

3.1 Lerner condition

The FOC $D'\mu + D\mu_p = 0$ implies

$$\mathcal{L} \equiv \frac{\mu}{p} = \frac{\mu_p}{\varepsilon_D}. \quad (9)$$

Without returns ($r \equiv 0, \mu_p = 1$), $\mathcal{L} = 1/\varepsilon_D$.

3.2 Pass-through of c and h

The monopolist's FOC is $f(p, c, h) \equiv D'(p) \mu + D(p) \mu_p = 0$. At the optimum, totally differentiate w.r.t. $x \in \{c, h\}$:

$$0 = \frac{\partial f}{\partial x} = \underbrace{D'\mu_x + D\mu_{px}}_{\text{direct effect}} + \underbrace{(D''\mu + 2D'\mu_p + D\mu_{pp})}_{\Delta} \rho_x,$$

since $\partial f / \partial p = 0$. Write $\Delta \equiv D''\mu + 2D'\mu_p + D\mu_{pp}$ with $\mu_{pp} = -2r'(p) - r''(p)(p + h)$. Hence

$$\rho_x = -\frac{D'\mu_x + D\mu_{px}}{\Delta}. \quad (10)$$

Production cost c . From $\mu = p - c - r(p)(p + h)$: $\mu_c = -1, \mu_{pc} = 0$. Then

$$\rho_c = -\frac{D'(-1) + D \cdot 0}{\Delta} = \frac{D'}{\Delta}. \quad (11)$$

Handling cost h . From μ : $\mu_h = -r(p)$ and $\mu_{ph} = \partial \mu_p / \partial h = -r'(p)$. Then

$$\rho_h = -\frac{D'(-r) + D(-r')}{\Delta} = \frac{D'r + Dr'}{\Delta}. \quad (12)$$

The $D'r$ term is production pass-through scaled by r ; the Dr' term comes from μ_{ph} . Dividing (12) by (11):

$$\frac{\rho_h}{\rho_c} = r + \frac{Dr'}{D'} = r(p) - \frac{pr'(p)}{\varepsilon_D(p)}, \quad (13)$$

using $D' = -D\mu_p/\mu$ from the FOC and $\varepsilon_D = -pD'/D$. When $r' = 0$, $\rho_h = r\rho_c$; when $r' > 0$, $\rho_h < r\rho_c$.

Example: $r(p) = \alpha$, $D(p) = a - bp$. Here $\mu_p = 1 - \alpha$, $\mu_{pp} = \mu_{ph} = 0$, and the FOC is $-b\mu + (a - bp)(1 - \alpha) = 0$. Differentiate w.r.t. c :

$$0 = -b(\mu_c + \mu_p \rho_c) + D'(1 - \alpha) \rho_c = b - 2b(1 - \alpha) \rho_c,$$

so $\rho_c = 1/[2(1 - \alpha)]$. Differentiate w.r.t. h :

$$0 = -b(\mu_h + \mu_p \rho_h) + D'(1 - \alpha) \rho_h = b\alpha - 2b(1 - \alpha) \rho_h,$$

so $\rho_h = \alpha/[2(1 - \alpha)] = \alpha \rho_c$.

3.3 Incidence

Envelope theorem: $d\pi/dc = -D$, $d\pi/dh = -Dr$; $dCS/dx = -D\rho_x$. Hence

$$I_c = \rho_c, \quad I_h = \frac{\rho_h}{r} = \rho_c \left(1 - \frac{p r'}{\varepsilon_D r}\right). \quad (14)$$

$I_h = \rho_c$ when $r' = 0$; $I_h < \rho_c$ when $r' > 0$.

3.4 Summary: monopoly

Object	Formula
Lerner	$\mathcal{L} = \mu_p/\varepsilon_D$
ρ_c	D'/Δ
ρ_h	$(D'r + Dr')/\Delta$
I_c	ρ_c
I_h	ρ_h/r

4 Perfect competition vs. monopoly

	Perfect competition	Monopoly
ρ_h/ρ_c	r	$r - pr'/\varepsilon_D$
I_c	$\rho_c/(1 - \mu_p \rho_c)$	ρ_c
I_h	I_c	ρ_h/r

References

References

[Weyl and Fabinger(2013)] Weyl, E. G. and M. Fabinger (2013). Pass-through as an economic tool: Principles of incidence under imperfect competition. *Journal of Political Economy*, 121(3):528–583.